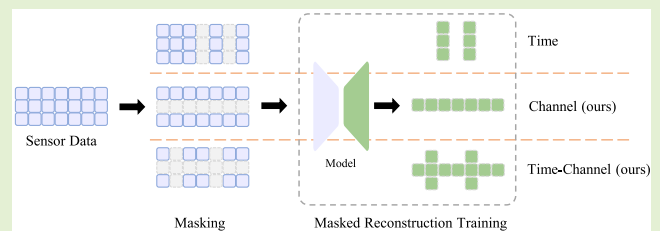


An Improved Masking Strategy for Self-Supervised Masked Reconstruction in Human Activity Recognition

Jinqiang Wang^{ID}, Wenxuan Cui^{ID}, Tao Zhu^{ID}, *Member, IEEE*,
Huansheng Ning^{ID}, *Senior Member, IEEE*, and Zhenyu Liu

Abstract—Masked reconstruction serves as a crucial pretext task for self-supervised learning, enabling the model to improve its feature extraction capabilities by reconstructing the masked segments from abundant unlabeled data. In the field of human activity recognition (HAR), this pretext task has traditionally employed a masking strategy that focuses on the time dimension. However, this strategy fails to fully leverage the inherent characteristics of wearable sensor data and overlooks the interchannel information coupling, thereby limiting its potential as an effective pretext task. To overcome these limitations, we propose a novel masking strategy called channel masking. This strategy involves masking the sensor data along the channel dimension, thereby encouraging the encoder to extract channel-specific features while performing the masked reconstruction task. Additionally, channel masking can be seamlessly integrated with masking strategies along the time dimension, motivating the self-supervised model to undertake the masked reconstruction task in both the time and channel dimensions. The integrated masking strategies are named time-channel masking and span-channel masking. Furthermore, we optimize the reconstruction loss function to incorporate the reconstruction loss in both the time and channel dimensions. We evaluate the proposed masking strategies on three public datasets, and the experimental results demonstrate that these strategies outperform prior approaches in both self-supervised and semisupervised scenarios.

Index Terms—Channel masking, human activity recognition (HAR), masked reconstruction, self-supervised, wearable sensors.



I. INTRODUCTION

THE proliferation of wearable devices has facilitated easy access to human activity data. This accessibility has sparked significant interest in human activity recognition (HAR) based on wearable devices, propelling the rapid

advancement of pervasive computing [1], [2]. This technology finds human-centered applications in diverse domains such as healthcare [3], [4], [5], sports assessment [6], [7], and smart homes [8], [9]. Deep learning methods [10], [11], [12], [13] have emerged as a pivotal driver in enhancing the performance of HAR when compared to traditional machine learning approaches [14], [15], [16]. These deep learning methods leverage deep neural networks to process sensor data and infer activity categories through classifiers. However, most deep learning methods follow the supervised learning paradigm, which requires abundant labeled data to train a model. Unfortunately, manually labeling sensor data proves to be an arduous and time-consuming task, presenting a formidable challenge that restricts the progress of HAR [17].

To address the challenge of the scarcity of labeled data in HAR, a promising solution has emerged: self-supervised learning [18]. This solution alleviates the issue of limited labeled data by formulating and addressing self-generated tasks on expansive datasets that do not necessitate human annotations. Capitalizing on the inherent structure and information within the data, self-supervised learning guides the learning process, empowering the encoder to capture valuable feature repre-

Manuscript received 13 March 2024; revised 15 April 2024; accepted 15 April 2024. Date of publication 24 April 2024; date of current version 31 May 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62006110 and in part by the Natural Science Foundation of Hunan Province under Grant 2023JJ30518. The associate editor coordinating the review of this article and approving it for publication was Dr. Chiman Kwan. (Jinqiang Wang and Wenxuan Cui contributed equally to this work.) (Corresponding authors: Tao Zhu; Zhenyu Liu.)

Jinqiang Wang is with the School of Computer Science, University of South China, Hengyang 421001, China, and also with the School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing 100083, China.

Wenxuan Cui, Tao Zhu, and Zhenyu Liu are with the School of Computer Science, University of South China, Hengyang 421001, China (e-mail: tzhu@usc.edu.cn; lzy@usc.edu.cn).

Huansheng Ning is with the School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing 100083, China (e-mail: ninghuansheng@ustb.edu.cn).

Data is available on-line at <https://github.com/diheal/channelMasking>. Digital Object Identifier 10.1109/JSEN.2024.3390755

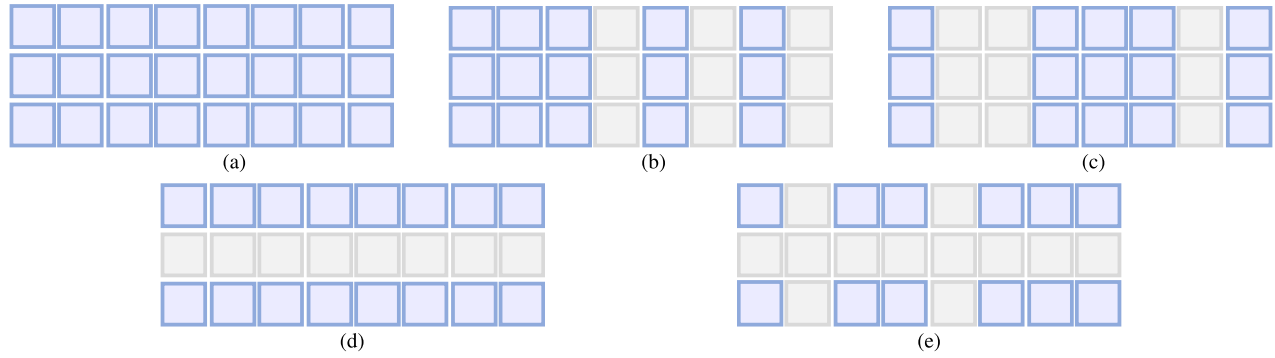


Fig. 1. Diagram of different masking strategies. The above example is a segment of sensor data with time on the horizontal axis and channels on the vertical axis. The gray block indicates the masked portion. (a) Raw data. (b) Time masking. (c) Span masking. (d) Channel masking. (e) Time-channel masking.

sentations. Currently, several self-supervised learning pretext tasks have been successfully applied to HAR, each yielding encouraging results. These tasks include augmentation method discrimination [19], contrastive learning [20], [21], [22], [23], data reconstruction [24], and masked reconstruction [25], [26].

Existing self-supervised models with masked reconstruction are randomly masking sensor data along the time dimension [25], [26]. As illustrated in Fig. 1(b), the sensor data are subject to random masking along the time axis. Additionally, Fig. 1(c) demonstrates the application of Span Masking, where contiguous segments of sensor data along the time axis are randomly masked. After masking the data, the self-supervised model endeavors to reconstruct the sensor data, incorporating the masked positions. Upon the completion of pretraining, the encoder of the self-supervised model is deployed for the downstream activity classification task.

However, the masking strategy along the time dimension does not comprehensively account for the characteristics of the sensor data and ignores the importance of the information interactions between the channels. Consequently, the encoder's feature extraction capability may be hindered, particularly when facing sensor channel anomalies.

To address this limitation, we propose a novel masking strategy called channel masking. As depicted in Fig. 1(d), sensor data are randomly masked along the channel dimension. This strategy can motivate self-supervised learning models to extract channel-dependent information. Furthermore, channel masking can be combined with masking strategies along the time dimension. The combined masking strategies are named time-channel masking and span-channel masking. Such masking strategies would empower the self-supervised model with the ability to model both time and channel dimensions and enhance the encoder's ability to extract semantic features. Finally, owing to the change in masking strategy, we optimized the previous loss function so that it can reconstruct the model to learn extracting both time-dependent and channel-dependent features.

We evaluate our masking strategies on three public datasets, namely USC-HAD, UCI-HAR, and MotionSense. Additionally, we assess the performance of these strategies

under two experimental scenarios: self-supervised learning, where the downstream task uses a large amount of labeled data, and semisupervised learning, where the downstream task uses a limited amount of labeled data. Experimental results show that our proposed strategies outperform previous masking strategies. Furthermore, this article explores the sensitivity of the hyperparameters of proposed masking strategies and the effect of whether the same mask position is used for the same training batch. Finally, this article evaluates channel masking in channel anomaly experiments, and its experimental results show that the self-supervised model with channel masking can significantly mitigate the channel anomaly problem. This article contributes the following.

- 1) This article considers first the importance of sensor channel-dependent features in the masked reconstruction task and proposes a new masking strategy called channel masking. It can be merged with masking strategies along the time dimension called time-channel masking and span-channel masking.
- 2) Our proposed masking strategies demonstrate superior performance over previous approaches on three public datasets and across two experimental scenarios. Additionally, feature map visualizations provide insights into the effectiveness of our strategy.
- 3) We optimize the reconstruction loss function to be compatible with both channel dimension and time dimension reconstruction losses.
- 4) This article demonstrates that our proposed masking strategy is effective in mitigating the channel anomaly problem compared to supervised learning.

The remainder of the article is structured as follows. Section II summarizes the related work; Section III describes in detail channel mask, time-channel masking, and span-channel masking, along with the optimized reconstruction loss functions. Section IV describes the experimental setup for evaluating our proposed method; Section V shows the main experimental results and discussion; Section VI provides further analysis and discussion on the details of the masking strategy; and Section VII summarizes the article and proposes new challenges based on the identified shortcomings.

II. RELATED WORK

A. Human Activity Recognition

HAR refers to the automatic identification of human actions by a computer system. These actions include walking, running, climbing stairs, lying down, and falling. This technology relies on the use of sensors to capture individuals' behavioral patterns. The collected sensor data are then processed by the computing system to enable activity recognition [27]. Accelerometers, gyroscopes, and magnetometers are commonly used sensors in this field [28]. To implement HAR techniques, various machine learning methods have been employed [14], [15], [16]. Subsequently, deep learning methods, which involve powerful feature extraction, are gradually gaining popularity in activity recognition. This is because traditional machine learning methods require manual feature selection and expert knowledge. Supervised deep learning methods have been proven to significantly improve the accuracy of activity recognition compared to traditional methods [10], [12], [29], [30], [31]. However, these supervised learning methods often require a large amount of annotated data for model training. Annotating data is a time-consuming, tedious, and error-prone process, thus posing a major challenge for HAR [17].

B. Self-Supervised Learning

Self-supervised learning, also known as pretraining, involves completing a pretext task by leveraging supervised signals generated in a nonhuman manner on a large amount of unlabeled data [18]. The use of pretrained encoders has been shown to enhance performance in downstream tasks, effectively addressing the challenge of limited labeled data availability. Several studies have explored the application of self-supervised learning in HAR. In one research [19], data transformation recognition was utilized as the pretext task to train the self-supervised model. In another study [24], multiple types of autoencoders were applied for the pretraining of activity data. The study [32] was the first to apply CPC [33] to the pretraining of activity data, achieving superior performance compared to previous approaches. Additionally, a study [20] introduced contrastive learning to HAR, investigating the impact of discriminative pretraining on performance. Another study [21] improved activity recognition performance by employing the transformer encoder for contrastive learning based on HAR. The study [22] proposed a sensor data augmentation method based on contrastive learning to enhance the performance of pretrained models. Finally, the study [23] improved pretraining performance by optimizing the negative selection method in contrastive learning.

Masked reconstruction is a widely pretext task in self-supervised learning, aimed at promoting the capture of contextual features by the model from the data. This task has applications in various domains. For example, in natural language processing, BERT [34] and GPT [35] utilize a pretext task of predicting masked words. Similarly, in computer vision, the MAE [36] pretext task involves reconstructing masked image patches.

Masked reconstruction tasks have proven effective in training self-supervised models for HAR [25]. The work [25]

utilized this task by randomly masking sensor data along the time axis and subsequently reconstructing these masked data. The default mask ratio in their study is set at 10%. The performance of activity recognition in this work surpassed that of other self-supervised paradigms at the time. Building upon this foundation, The work [26] introduces the idea of Span Masking [37], which masks continuous segments of sensor data along the time axis and then reconstructing these masked segments. In their experimental setup, the default mask ratio is set to 15%. This work further demonstrates the potential of masked reconstruction tasks. Nevertheless, the masking strategies of the above works primarily target on the time dimension of sensor data, ignoring the sensor channel dimension that contains abundant activity information. Hence, there is a pressing need for further exploration into masking strategies tailored for the channel dimension.

III. METHOD

To fully harness the potential of self-supervised learning models utilizing masked reconstruction and to comprehensively account for the intrinsic characteristics of sensor data, we introduce an innovative masking strategy for sensor data, called channel masking. The procedural steps for implementing this masking strategy are delineated as follows.

A continuous segment of sensor data can be expressed in the form shown in (1), where N is the length of time, K denotes the number of channels, and x_i^j denotes the data of the j th channel at the i th moment. The channel masking operation is shown in (2), where C is a randomly taken set of channel indices, and $\mathbb{1}_{[j \in C]}$ is an indicator function, which is equal to 1 when the expression in $[\cdot]$ is true and 0 otherwise. The channel masking schematic is shown in Fig. 1(d)

$$x_i^j; \quad i = 1, 2, \dots, N; \quad j = 1, 2, \dots, K \quad (1)$$

$$\tilde{x}_i^j = (1 - \mathbb{1}_{[j \in C]}) \cdot x_i^j; \quad i = 1, 2, \dots, N, \quad j = 1, 2, \dots, K. \quad (2)$$

Channel masking can be combined with masking strategies along the time axis as shown in (3), where T is the set of indices of the time dimension to be masked. The data are masked along both time and channel dimensions. If T is generated by randomization [25], the combined masking strategy is named time-channel masking. If T is generated by span masking [26], [37], the combined masking strategy is called span-channel masking

$$\tilde{x}_i^j = (1 - \mathbb{1}_{[j \in C]}) \cdot \tilde{x}_i^j = (1 - \mathbb{1}_{[i \in T][j \in C]}) \cdot x_i^j \\ i = 1, 2, \dots, N; \quad j = 1, 2, \dots, K. \quad (3)$$

The above proposed masking strategies will be applied as the training process shown in Fig. 2. This framework is based on the improvement of the mask reconstruction framework, and we focus on optimizing the data masking strategy. Given the incorporation of channel-related information within the masking strategies, the conventional reconstruction loss function becomes inapplicable to this new framework. Consequently, it becomes imperative to revise the reconstruction loss function, accommodating the complexities introduced by both time- and channel-based masking strategies.

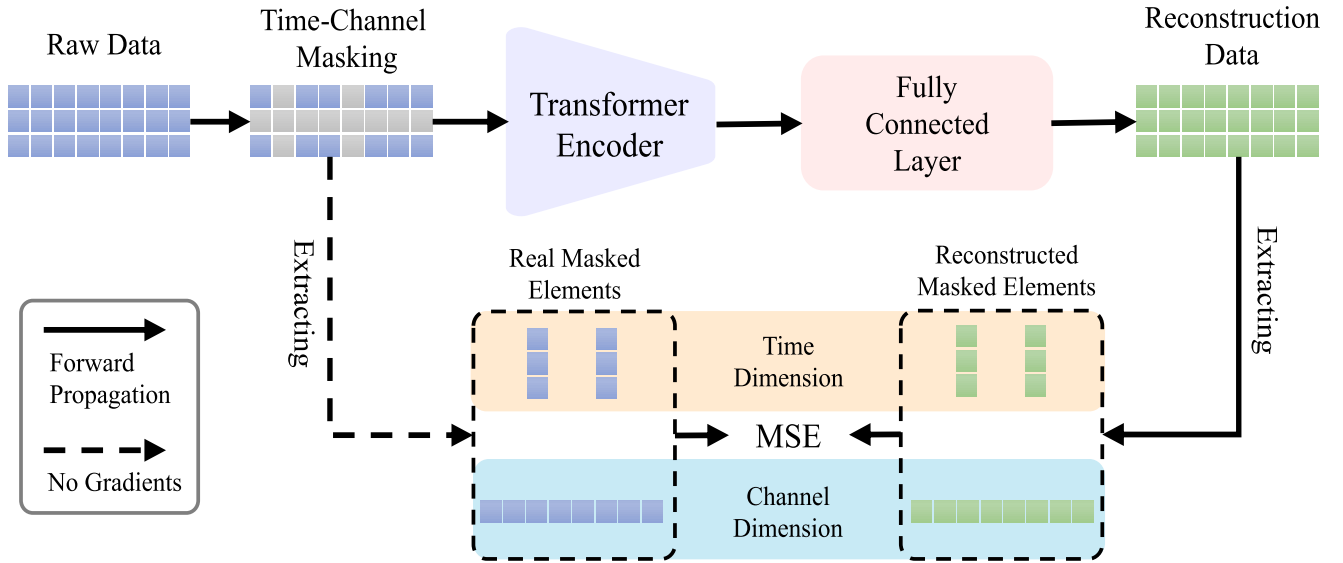


Fig. 2. Self-supervised learning framework of masked reconstruction using time-channel masking. The raw data are masked by masking strategies. The resulting masked data subsequently serve as input to the encoder, facilitating the extraction of intermediate features. These intermediary features are then directed into a fully connected layer, tasked with the reconstitution of the raw data. Following this stage, the difference in the masked positions between the reconstructed data and the raw data denoted as the reconstruction loss is computed. Subsequently, during the downstream task, only the encoder with frozen parameters is used.

The original mask reconstruction loss function is shown in (4). Herein, X_T denotes the mask data along the time dimension in the original data, X'_T denotes the mask data along the time dimension in the reconstructed data, and T is the set of masked time indexes. The term “MSE” signifies the mean square error loss. The loss function corresponding to our proposed channel masking strategy is shown in (5). Similar to (4), X_C denotes the mask data along the channel dimension in the original data, X'_C denotes the mask data along the channel dimension in the reconstructed data, and C is the set of masked channel indexes.

To incorporate both the time masking and channel masking strategies into the loss function, we merge L_T and L_C by weighting as outlined in (6). The parameter $\alpha \in [0, 1]$ is introduced to measure the relative significance assigned to the time and channel dimensions within the model. Unless otherwise specified, α is set to 0.5

$$L_T = \text{MSE}(X_T, X'_T) \quad (4)$$

$$L_C = \text{MSE}(X_C, X'_C) \quad (5)$$

$$L = \alpha \cdot L_T + (1 - \alpha) \cdot L_C. \quad (6)$$

IV. EXPERIMENTAL SETUP

A. Datasets

The USC-HAD [38] activity recognition dataset comprises accelerometer and gyroscope data extracted from 14 participants who engaged in 12 daily activities. These activities encompass walking forward, walking left, walking right, walking upstairs, walking downstairs, running, jumping, sitting, standing, lying down, and riding up and down in an elevator. The accelerometer and gyroscope data were captured utilizing a MotionNode device positioned in the volunteer’s right hip. Subsequently, these data were transmitted to a laptop computer via wired connections. All data were collected at a 100 Hz sampling rate. Consistent with established practices [21], [25],

the data were segmented using a 1 s sliding window with a 50% overlap, resulting in a total of 54 619 samples. The distribution of the various action classes was as follows: walking forward: 7488 samples, walking left: 5150 samples, walking right: 5256 samples, walking upstairs: 4096 samples, walking downstairs: 3808 samples, running forward: 3390 samples, jumping up: 2002 samples, sitting: 5090 samples, standing: 4580 samples, sleeping: 7360 samples, elevator up: 3204 samples, and elevator down: 3195 samples.

The UCI-HAR [16] activity recognition dataset encompasses accelerometer and gyroscope data that were collected from 30 volunteers aged 19–48 years. These individuals performed six fundamental activities, namely standing, sitting, lying down, walking, ascending, and descending stairs. Specifically, these data were collected utilizing a smartphone (Samsung Galaxy S II) equipped with an embedded accelerometer and gyroscope. These sensors were affixed to the waist of each volunteer, enabling the capture of three-axis linear acceleration and three-axis angular velocity. This acquisition was conducted consistently at a sampling rate of 50 Hz. Based on the experience of [23], [40], this dataset was segmented using a sliding window of 128 reads with 50% overlap, yielding 10 299 samples. The dataset was composed as follows: walking: 1722 samples, walking upstairs: 1544 samples, walking downstairs: 1406 samples, sitting: 1777 samples, standing: 1906 samples, and laying: 1944 samples.

The MotionSense [39] activity recognition dataset is a collection of accelerometer data and gyroscope data from 24 volunteers of different genders, ages, weights, and heights. These volunteers undertook six basic activities: going downstairs, going upstairs, walking, jogging, sitting, and standing. The specific collection process involved placing a smartphone (iPhone 6s) in the front pocket of the volunteers. Subsequently, the accelerometer and gyroscope data were systematically collected via the Core Motion framework on iOS devices

using SensingKit. The sampling frequency was steadfastly maintained at 50 Hz throughout the entire data collection process. In line with prior studies [21], [25], the data were segmented using a 1 s sliding window with a 50% overlap, resulting in a total of 55 967 samples. The distribution of the classes was as follows: downstairs: 13 477 samples, upstairs: 12 184 samples, sitting: 13 665 samples, standing: 6184 samples, walking: 5162 samples, and jogging: 5295 samples.

B. Self-Supervised Experimental Settings

In consonance with prior studies [23], [25], the data from subjects 11–12 in the USC-HAD dataset were designated as the validation set, subjects 13–14 were designated as the test set, and the rest of the subjects were designated as the training set. Building on the established practices of [23] and [40], 20% of the subjects in the UCI-HAR and MotionSense were randomly selected as the test set, 20% of the subjects in the rest of the data were used as the validation set, and the remaining subjects' data were used as the training set.

The structure of the self-supervised model is as follows.

- 1) *Encoder*: The initial component requires a single CNN layer with a convolution kernel size of (1, 1), generating 128 output channels. Subsequently, this is succeeded by a transformer–encoder network which uses cosine positional coding, attention block is three layers, and the number of heads of multi-head attention is 4.
- 2) *Fully Connected Layer*: It consists of three layers of MLP with the number of neurons accordingly by 256, 128, and the number of data channels. Between every two layers, there exists batch normalization, ReLU, and dropout with a dropout rate of 0.1.

The pretraining procedure is to train the self-supervised model for 150 epochs. Each epoch employs a batch size of 256, utilizing the Adam [41] optimizer with an initial learning rate of $1e-3$. The loss function presented in (4) is applied to the time dimension masking strategy, while the loss function elucidated in (5) is employed for the channel dimension masking strategy. Furthermore, the loss function expounded in (6) is used for the integrated masking strategy. Note that a training batch uses the same mask position generated randomly.

For the downstream task, the encoder acquired from the pretraining phase is employed with all layers frozen. Subsequently, the fully connected layers are randomly initialized, with the number of neurons in the last layer matching the number of activity categories. The model is trained on a labeled training set, and the performance of the pretrained model is evaluated on a test set. The mean $F1$ -score [42] was used as the evaluation metric. The model was trained for 100 epochs with a batch size of 1024. Adam [41] optimizer with an initial learning rate of $1e-3$ was used to optimize the model. Cross entropy loss is used as the loss function.

Since the test sets of the UCI-HAR and MotionSense datasets were obtained by random division, we sampled the datasets five times and trained them separately, and the final results were averaged. In this article, each experimental result is obtained by conducting five individual experiments and then calculating the average value.

C. Semisupervised Experimental Settings

In the semisupervised experimental setup, the process and parameters of pretraining the model are the same as in the self-supervised experimental setup. In the downstream tasks, in order to reflect the shortage of labeled data, k labeled samples of each category are extracted from the training set obtained by the dataset division method in Section IV-B as a semisupervised training set, where $k \in \{1, 2, 5, 10, 25, 50, 100\}$. The validation set and test set settings are identical to the self-supervised setup. The rest of the experimental setup follows the self-supervised experimental setup unless otherwise stated.

V. EXPERIMENTAL RESULT

A. Self-Supervised Experiment

In this section, we conduct an empirical evaluation to assess the efficacy of self-supervised models employing various masking strategies within the context of an activity recognition task. Specifically, we consider masking ratios of 10% for time masking and 15% for span masking based on the experience of the work [25], [26]. Our proposed channel masking introduces a 50% masking ratio, meaning that for USC-HAD and MotionSense datasets, three randomly chosen channels are masked, and for UCI-HAR, five channels are masked. For time-channel masking, the ratio of time dimension masking is 10% and the number of channel masks is 2 on all datasets. For span-channel masking, the ratio of time dimension masking is 15% and the number of channel masks is 2 on all datasets. Here, the purpose of combining masks is to explore whether the performance of the model is improved by adding channel masks based on time masks. The performance of the supervised learning model is also compared, which differs from the self-supervised learning model in that the encoder is randomly initialized instead of freezing the pretraining parameters.

Given the scarcity of masking strategies in HAR, we evaluate point masking and patch masking, previously utilized in diverse domains [36]. Point masking randomly masks sensor data points, while patch masking first divides sensor data into patches and randomly masks these patches. Following prior research [36], both strategies maintain a 75% masking ratio across all datasets. The patch size is set to two along the sample channel dimension and 6.25% along the time dimension. To comprehensively evaluate our approach, this section introduces the new metrics: accuracy, precision, and recall.

The experimental results presented in Table I show that our proposed masking strategies exhibit superior performance compared to the previous strategies across three distinct datasets and four evaluation metrics. This observation demonstrates that prompting the self-supervised learning model to consider channel-related information can improve its performance on activity recognition tasks. On the three datasets, the difference between the performance of our proposed optimal masking strategy and the performance of the optimal previous masking strategy is 7.41, 1.54, and 1.70 under the $F1$ -score metric, respectively. These differences suggest that the performance of the self-supervised learning framework based on masked reconstruction is highly dependent on the

choice of masking strategy. In the comparative analysis with supervised learning, the self-supervised learning model significantly beats supervised learning on the USC-HAD dataset only and insignificantly beats the supervised learning model on the UCI-HAR and MotionSense datasets. The difference in performance may be due to the supervised learning model perform better on the UCI-HAR and MotionSense datasets, which limited the potential of the self-supervised learning model. Conversely, the USC-HAD dataset reveals suboptimal performance by the supervised learning model, affording ample room for improvement by the self-supervised paradigm.

B. Semisupervised Experiment

Here, we explore the impact of different masking strategies for the self-supervised learning framework in the setting of limited labeled data. The experimental setup here follows the semisupervised experimental setup. The supervised learning benchmarks are also compared with the same model structure as presented in the downstream task of the self-supervised model, with the difference that its parameters are randomly initialized. The experimental results are shown in Fig. 3, where point masking and patch masking are defined in Section V-A.

On the USC-HAD and UCI-HAR datasets, experimental results show that our proposed masking strategies consistently surpass previous masking strategies across all labeling ratios. The associated confidence intervals conspicuously establish its statistically significant superiority. In the MotionSense dataset experiments, only point masking exhibits comparable performance to our proposed masking strategy, while all other strategies are significantly inferior to ours. This observation suggests that the self-supervised learning model considering channel-related information can improve activity classification performance in a restricted labeling environment. Notably, the self-supervised model, employing our devised masking strategies, outperforms the supervised learning benchmark. Particularly, for the USC-HAD dataset, it can be seen by the confidence intervals that our proposed masking strategies significantly outperform supervised learning models, while the prior masking strategies significantly trail behind the supervised learning models. This once again proves the importance of our proposed strategies. On the UCI-HAR and MotionSense datasets, our proposed masking strategies exhibit marginal and statistically insignificant performance improvements over the supervised learning model across the majority of experimental setups. Conversely, our approaches are insignificantly worse than the supervised learning model for a small number of setups. This observation could be attributed to the intrinsic ease of training the UCI-HAR and MotionSense datasets, which potentially constrains the potential improvement space for the self-supervised learning model. Overall, in a limited labeled data setting, the self-supervised learning model exhibits more pronounced improvements relative to the supervised learning model, as opposed to situations with ample labeled data.

C. t-SNE Visualization

To visualize the performance of various masking strategies, we conducted dimensionality reduction on the output of the

pretrained model employing the t-SNE method. We utilized the unlabeled test set from UCI-HAR as input samples, ensuring that no masking operations were applied and that these samples were not involved in the pretraining phase. Then, we reduced the output of the pretrained model to a 2-D space and depicted the results through scatterplots. Note that the pretrained model has not been injected with any labeling information. The visualization results are shown in Fig. 4.

It can be clearly seen that our method can pull in samples of the same class and push away samples of different classes under the latent space compared to other methods. Notably, samples annotated black, blue, and green can be better distinguished from other classes. This observation significantly demonstrates that our proposed masking strategy can enhance the ability of pretrained models to extract the domain-invariant features of the sample and better serve the downstream tasks.

VI. DISCUSSION

This section is dedicated to investigating the impact of the hyperparameters associated with the proposed masking strategies on model performance. A pretraining trick tailored for a self-supervised model based on masked reconstruction is also explored. According to previous experience [23], the smaller the number of labels, the higher the parameter sensitivity and the more intuitive the corresponding performance difference. Thus, the experimental setup in this section follows the semisupervised experimental setup, with the only difference being that in the downstream task, the sampling size of each class randomly sampled from the training set is fixed to 10.

A. Masking Ratio

The masking ratio exerts a direct influence on the complexity of the reconstruction task. The larger the masking ratio is, the less valid information is contained in the masked data, and the more difficult the reconstruction task is. For this reason, the effect of the masking ratio on the performance of the self-supervised model is explored. Here, the time and channel dimensions are explored separately, and the experimental results are shown in Tables II and III.

Time masking is discussed first. As the degree of masking amplifies, there is a concurrent enhancement observed in the overall model performance across all three datasets. Notably, there are only transient and localized performance dips. On all three datasets, the most effective ratio of masking is determined to be 90%. This revelation diverges from established conventions rooted in earlier research, where prevalent practice involved the utilization of default masking ratios of either 10% or 15%. The empirical findings demonstrate that our proposed self-supervised learning method, which is based on mask reconstruction, can effectively adapt to large-scale temporal masks and achieve superior performance compared to small-scale temporal masks.

Channel masking is discussed second. On both the USC-HAD and MotionSense datasets, the model performance is increasing and then decreases as the masking number increases. The best model performance corresponds to a masking number of 3, which is exactly half the number

TABLE I

PERFORMANCE OF PRETRAINED MODELS USING VARIOUS MASKING STRATEGIES ON AN HAR TASK. THE EVALUATION METRIC IS $F1$ (MEAN $F1$ -SCORE), ACC (ACCURACY), P (PRECISION), AND R (RECALL). "SUPERVISED" DENOTES A SUPERVISED LEARNING MODEL. THE REST OF THE EXPERIMENTAL RESULTS FOLLOW THE SELF-SUPERVISED EXPERIMENTAL SETUP

	USC-HAD				UCI-HAR				MotionSense			
	F1	ACC	P	R	F1	ACC	P	R	F1	ACC	P	R
Supervised	56.55	65.08	67.14	63.11	92.19	92.15	92.29	92.15	83.55	85.13	86.01	84.59
Time Masking	52.55	60.17	65.42	57.18	79.60	80.06	81.07	79.57	82.72	82.87	83.21	82.45
Span Masking	51.42	59.98	63.69	58.14	89.56	89.69	89.94	89.43	81.7	83.77	85.17	81.91
Point Masking ¹	56.91	63.01	68.57	59.84	92.17	92.29	92.91	91.57	81.68	84.16	84.71	83.87
Patch Masking ²	23.78	24.25	24.87	21.69	86.98	88.66	90.16	86.34	78.56	79.55	81.70	77.23
Channel Masking(ours)	62.83	68.56	70.74	67.33	93.08	92.89	93.21	92.45	84.42	84.35	85.51	82.35
Time-Channel Masking(ours)	64.32	70.47	76.29	67.12	93.24	93.08	93.17	92.99	83.92	83.88	84.65	82.37
Span-Channel Masking(ours)	58.88	65.99	72.51	60.10	93.71	93.29	95.28	92.86	83.73	83.85	84.18	83.58

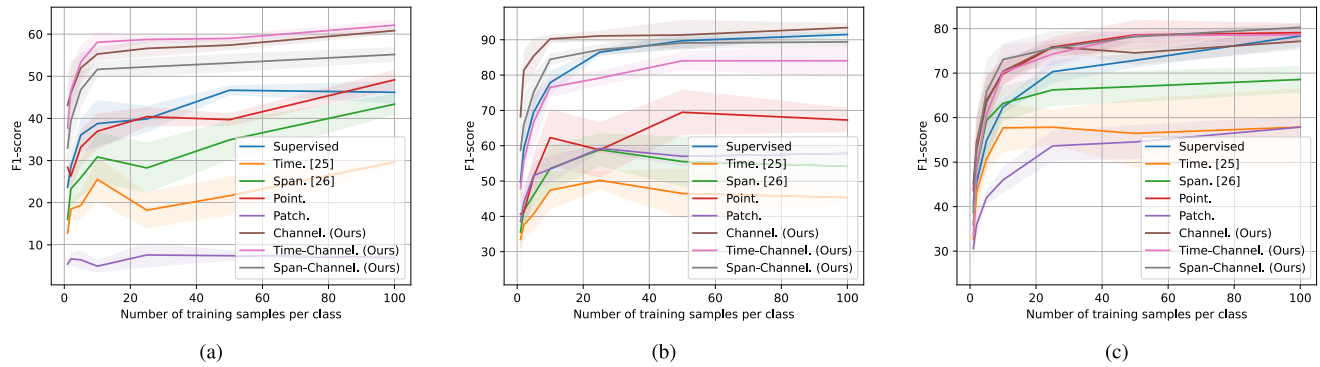


Fig. 3. Performance of self-supervised learning models using different masking strategies for activity classification in limited labeled data. The evaluation metric is $F1$ -score. All results are obtained by averaging five experimental results, and the shaded areas denote 95% confidence intervals. (a) USC-HAD. (b) UCI-HAR. (c) MotionSense.

TABLE II

CLASSIFICATION PERFORMANCE OF SELF-SUPERVISED LEARNING MODELS USING DIFFERENT TIME MASKING RATIOS. THE $F1$ -SCORE SERVES AS THE EVALUATION METRIC, WITH THE DESIGNATED MASKING STRATEGY BEING TIME MASKING. THE "MASKING RATIOS" DENOTE THE PROPORTION OF MASKED SENSOR DATA ALONG THE TIME AXIS

Masking Ratios	USC-HAD	UCI-HAR	MotionSense
5%	7.15	41.95	52.22
10%	22.17	32.48	49.46
20%	28.35	50.12	58.40
30%	17.74	51.11	57.80
50%	19.71	56.69	65.67
70%	24.86	61.53	64.79
90%	39.31	65.59	70.07

of sensor data channels. On the UCI-HAR dataset, optimal performance aligns with a masking number of 4, akin to the performance at a masking number of 5. Notably, the UCI-HAR dataset has a data channel number of 9, and the optimal masking number is also right around half of the number of data channels. The above discussion leads to the conclusion that leveraging channel masking with cardinality equal to half the data channels remarkably enhances self-supervised learning models rooted in masked reconstruction. This conclusion doubles as a practical guideline for training other sensor datasets. Finally, by comparing Tables II and III,

TABLE III

CLASSIFICATION PERFORMANCE OF SELF-SUPERVISED LEARNING MODELS EMPLOYING VARIED CHANNEL MASKING NUMBERS. THE EVALUATION METRIC EMPLOYED IS THE $F1$ -SCORE, WITH THE SELECTED MASKING STRATEGY BEING CHANNEL MASKING. THE "MASKING NUMBERS" SIGNIFIES THE NUMBER OF MASKING SENSOR DATA CHANNELS. THE NUMBER OF DATA CHANNELS IS SIX FOR THE USC-HAD AND MOTIONSENSE DATASETS AND NINE FOR THE UCI-HAR DATASET

Masking Numbers	USC-HAD	UCI-HAR	MotionSense
1	43.74	60.85	70.27
2	45.66	70.37	69.61
3	57.21	71.90	71.79
4	51.23	78.81	68.70
5	37.24	78.53	60.44

it is found that the peak performance of the model with channel masking outperforms the best performance of the model with time masking. The findings again prove that the self-supervised learning model that considers channel-related information outperforms the self-supervised learning model that considers time-related information.

B. Hyperparameter α

Here, we delve into the significance of time reconstruction loss and channel reconstruction loss within the time-channel masking approach. Specifically, we analyze the impact of parameter α in (6) on model performance. To better understand

¹The method references pixel point masking in computer vision.

²Similar to 1, the method references patch masking in computer vision.

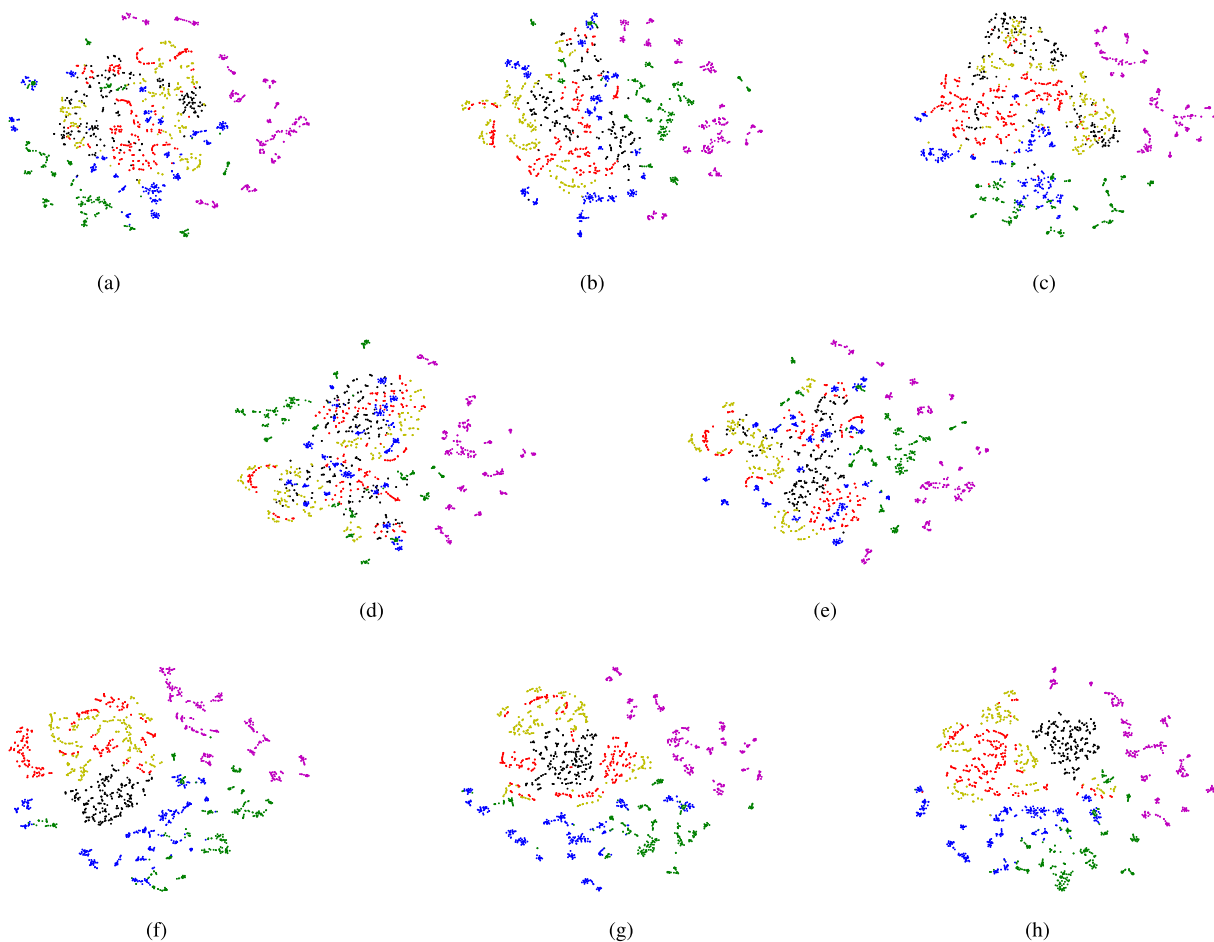


Fig. 4. Visualization of pretrained model outputs employing diverse masking strategies. The input samples are derived from the UCI-HAR test set and did not involve any masking operations. Activity categories are denoted by distinct colors: “Walking” in red, “Upstairs” in yellow, “Downstairs” in black, “Sitting” in green, “Standing” in blue, and “Laying” in magenta. (a) Model’s parameters are randomly initialized and have not undergone any training. (b) Point masking. (c) Patch masking. (d) Time masking. (e) Span masking. (f) Channel masking (ours). (g) Time-channel masking (ours). (h) Span-channel masking (ours).

TABLE IV
EFFECT OF VARYING HYPERPARAMETERS α ON MODEL PERFORMANCE USING THE TIME-CHANNEL MASKING STRATEGY. DECREASING α CORRESPONDS TO A LOWER PERCENTAGE OF TIME RECONSTRUCTION LOSS. THE EVALUATION METRIC IS $F1$ -SCORE

α	USC-HAD	UCI-HAR	MotionSense
0.1	45.50	61.13	70.52
0.3	48.45	70.60	70.79
0.5	47.71	77.70	69.42
0.7	44.41	71.44	44.41
0.9	46.57	68.25	46.57

the importance of these reconstruction losses, we maintain similar mask ratios for both time and channel aspects. For the USC-HAD and MotionSense datasets, the channel masking number was 1 (16.7%) and the time masking ratio was 17%. Likewise, for the UCI-HAR dataset, the channel masking number is 2 (22.2%) and the time masking ratio is 22%. Detailed experimental results are presented in Table IV.

The experimental results show that in the USC-HAD and MotionSense datasets, the best performance corresponds to

an α of 0.3, meaning that the model performs best when the channel reconstruction loss accounts for a large portion of the model. This empirical validation further emphasizes the importance of channel-dependence information for self-supervised learning models based on masked reconstruction. On the UCI-HAR dataset, the best performance corresponds to an α of 0.5, namely, the model performs best when the channel reconstruction loss and the time reconstruction loss account for the same percentage. With the above analysis, the value of the hyperparameter α can be assigned to be slightly less than 0.5 when pretraining a new dataset. In other words, the weight of the channel reconstruction loss is set to be slightly higher than the time reconstruction loss.

C. Sensor Channel Anomaly Issues

Here, we explore whether the self-supervised learning model after pretraining using the channel masking strategy has the ability to handle data channel anomalies. To keep the generality intact, the masking number for channel masking is set to 3 in the pretraining. In this task, pretrained models are trained with normal data and evaluated using channel anomaly data. Furthermore, a comparative analysis is conducted against

TABLE V

EFFECT OF VARYING HYPERPARAMETERS α ON MODEL PERFORMANCE USING THE TIME-CHANNEL MASKING STRATEGY. DECREASING α CORRESPONDS TO A LOWER PERCENTAGE OF TIME RECONSTRUCTION LOSS. THE EVALUATION METRIC IS $F1$ -SCORE

Masking Number	Paradigms	USC-HAD	UCI-HAR	MotionSense
1	Supervised	45.87	73.87	51.25
	Self-supervised	60.40	79.53	75.07
2	Supervised	37.63	68.24	44.32
	Self-supervised	47.83	77.67	62.91
3	Supervised	22.04	55.37	32.46
	Self-supervised	18.31	65.86	34.59

supervised learning benchmarks, the architectural description of which is presented in Section V. The experimental results are presented in Table V.

Experimental results show that the self-supervised learning model with channel masking surpasses the supervised learning model across the UCI-HAR and MotionSense datasets. This demonstrates that the model employing masked reconstruction of channel dimensions as a pretext task can improve the processing ability for channel anomaly data. On the USC-HAD dataset, the self-supervised learning model outperforms the supervised learning model when the masking number is 1 and 3, while inferior to the supervised model when the number of masks is 5. The divergence could potentially be attributed to the number of masked channels used in the pretraining is 3, which caused the model to overfit to this masking number. It is worth noting that pretraining was done using a masking channel number of 3, but when evaluated on anomalous data with a masking number of 1, it outperforms supervised learning on all three datasets, which suggests a certain degree of generalizability of the channel masking strategy.

D. Pretraining Trick

Here, we explore a trick in pretraining a self-supervised learning model based on masked reconstruction that uses the same mask position for the same batch of data. Whereas the previous method has different mask positions between each sample within the same training batch. Now the pretraining uses two tricks, same position masking and different position masking, for the same batch of samples, with time-channel masking as the masking strategy and then evaluates the performance of the two tricks in the downstream activity classification task separately. Detailed results can be found in Table VI.

Empirical findings reveal that across the three datasets, the performance of the model using the same mask position for the same batch is higher than that of the model using different mask positions. The differences between the two tricks on the three datasets are 13.62, 12.65, and 9.65, respectively, demonstrating that the trick of the same mask position can substantially improve the performance of the pretrained mode. This trick can also be utilized in subsequent self-supervised learning methods that conform to masked reconstruction.

TABLE VI

EFFECT OF THE SAME BATCH OF DATA USING THE SAME MASKING AND DIFFERENT MASKING TRICKS ON MODEL PERFORMANCE. TIME-CHANNEL MASKING WAS SET ON THE USC-HAD DATASET WITH A TIME MASKING RATIO OF 20 AND A CHANNEL MASKING NUMBER OF 1. ON THE UCI-HAR AND MOTIONSENSE DATASETS, THE TIME MASKING RATIO WAS SET TO 10 AND THE NUMBER OF CHANNELS MASKED WAS 2. THE ABOVE MASKING RATIOS WERE NOT INTENTIONALLY SET

Tricks	USC-HAD	UCI-HAR	MotionSense
Different.	36.37	62.33	62.38
Same. (ours)	49.99	74.98	72.03

VII. CONCLUSION

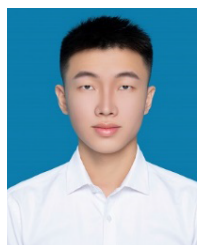
To maximize the potential of self-supervised learning models based on mask reconstruction for HAR tasks, this article proposes a new masking strategy channel masking, coupled with a time dimension masking strategy, yielding time-channel masking and span-channel masking. Proposed strategies are rigorously evaluated on three public datasets, encompassing both self-supervised and semisupervised experimental scenarios. Results substantiate the superiority of the proposed masking strategies over preceding strategies, underscoring the pivotal role of channel-dependent information. Specifically, our proposed strategies yield improvements of 7.41 (USC-HAD), 1.54 (UCI-HAR), and 1.70 (MotionSense) in $F1$ -score metrics compared to previous strategies in the self-supervised experiments. Furthermore, the proposed channel masking strategy is evaluated in sensor channel anomaly experiments, affirming channel masking can enhance the model's capability to handle data channel anomalies. Finally, the hyperparameters of the proposed masking strategies and a training trick are explored, and relevant recommendations are given based on the experimental results.

In addition, our study does exhibit certain limitations. When confronted with a substantial volume of annotated data, the self-supervised model employing proposed masking strategies does evince certain disparities when juxtaposed with its supervised counterpart. This gap is also a problem shared by other self-supervised learning models. We will bridge this gap by improving the reconstruction effect and eliciting other supervised signals in future work.

REFERENCES

- [1] L. M. Dang, K. Min, H. Wang, M. J. Piran, C. H. Lee, and H. Moon, "Sensor-based and vision-based human activity recognition: A comprehensive survey," *Pattern Recognit.*, vol. 108, Dec. 2020, Art. no. 107561.

- [2] Z. Zhang, R. Yin, and H. Ning, "Internet of Brain, thought, thinking, and creation," *Chin. J. Electron.*, vol. 31, no. 6, pp. 1025–1042, 2022.
- [3] S. Javaid, S. Zeadally, H. Fahim, and B. He, "Medical sensors and their integration in wireless body area networks for pervasive healthcare delivery: A review," *IEEE Sensors J.*, vol. 22, no. 5, pp. 3860–3877, Mar. 2022.
- [4] Y. Zhao et al., "Development of flexible electronic biosensors for healthcare engineering," *IEEE Sensors J.*, vol. 24, no. 8, pp. 11998–12016, Apr. 2024.
- [5] X. Zhou, W. Liang, K. I. Wang, H. Wang, L. T. Yang, and Q. Jin, "Deep-learning-enhanced human activity recognition for Internet of Healthcare Things," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6429–6438, Jul. 2020.
- [6] A. L. Clouthier, G. B. Ross, and R. B. Graham, "Sensor data required for automatic recognition of athletic tasks using deep neural networks," *Front. Bioeng. Biotechnol.*, vol. 7, p. 473, Jan. 2020.
- [7] Y. Ke, "Location and tracking mode of sports rehabilitation training with self-powered sensors based on particle swarm optimization algorithm," *IEEE Sensors J.*, vol. 23, no. 18, pp. 20894–20903, Sep. 2023.
- [8] S. K. Hiremath, Y. Nishimura, S. Chernova, and T. Plötz, "Bootstrapping human activity recognition systems for smart homes from scratch," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 6, no. 3, pp. 1–27, Sep. 2022.
- [9] L. Babangida, T. Perumal, N. Mustapha, and R. Yaakob, "Internet of Things (IoT) based activity recognition strategies in smart homes: A review," *IEEE Sensors J.*, vol. 22, no. 9, pp. 8327–8336, May 2022.
- [10] Y. Tang, L. Zhang, F. Min, and J. He, "Multiscale deep feature learning for human activity recognition using wearable sensors," *IEEE Trans. Ind. Electron.*, vol. 70, no. 2, pp. 2106–2116, Feb. 2023.
- [11] S. K. Challa, A. Kumar, and V. B. Semwal, "A multibranch CNN-BiLSTM model for human activity recognition using wearable sensor data," *Vis. Comput.*, vol. 38, no. 12, pp. 4095–4109, Dec. 2022.
- [12] J. Stuchbury-Wass, A. Ferlini, and C. Mascolo, "Multimodal attention networks for human activity recognition from earable devices," in *Proc. ACM Int. Joint Conf. Pervasive Ubiquitous Comput.*, Sep. 2022, pp. 258–260.
- [13] J. Feng and M. Gou, "A progressive region-focused network for fine-grained human behavior recognition," *Hum.-Centric Comput. Inf. Sci.*, vol. 13, pp. 1–18, Jun. 2023.
- [14] L. Fan, Z. Wang, and H. Wang, "Human activity recognition model based on decision tree," in *Proc. Int. Conf. Adv. Cloud Big Data*, Dec. 2013, pp. 64–68.
- [15] L. M. Rodrigues and M. Mestria, "Classification methods based on Bayes and neural networks for human activity recognition," in *Proc. 12th Int. Conf. Natural Comput., Fuzzy Syst. Knowl. Discovery*, Aug. 2016, pp. 1141–1146.
- [16] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "Human activity recognition on smartphones using a multiclass hardware-friendly support vector machine," in *Proc. Int. Workshop Ambient Assist. Living*. Cham, Switzerland: Springer, 2012, pp. 216–223.
- [17] K. Chen, D. Zhang, L. Yao, B. Guo, Z. Yu, and Y. Liu, "Deep learning for sensor-based human activity recognition: Overview, challenges, and opportunities," *ACM Comput. Surv.*, vol. 54, no. 4, pp. 1–40, 2021.
- [18] X. Liu et al., "Self-supervised learning: Generative or contrastive," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 1, pp. 857–876, Nov. 2021.
- [19] A. Saeed, T. Ozcelebi, and J. Lukkien, "Multi-task self-supervised learning for human activity detection," *Proc. ACM Interact., Mobile, Wearable Ubiquitous Technol.*, vol. 3, no. 2, pp. 1–30, Jun. 2019.
- [20] C. Ian Tang, I. Perez-Pozuelo, D. Spathis, and C. Mascolo, "Exploring contrastive learning in human activity recognition for healthcare," 2020, *arXiv:2011.11542*.
- [21] B. Khaertdinov, E. Ghaleb, and S. Asteriadis, "Contrastive self-supervised learning for sensor-based human activity recognition," in *Proc. IEEE Int. Joint Conf. Biometrics (IJCB)*, Aug. 2021, pp. 1–8.
- [22] J. Wang, T. Zhu, J. Gan, L. L. Chen, H. Ning, and Y. Wan, "Sensor data augmentation by resampling in contrastive learning for human activity recognition," *IEEE Sensors J.*, vol. 22, no. 23, pp. 22994–23008, Dec. 2022.
- [23] J. Wang, T. Zhu, L. Chen, H. Ning, and Y. Wan, "Negative selection by clustering for contrastive learning in human activity recognition," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10833–10844, Jun. 2023.
- [24] H. Haresamudram, D. V. Anderson, and T. Plötz, "On the role of features in human activity recognition," in *Proc. 23rd Int. Symp. Wearable Comput.*, Sep. 2019, pp. 78–88.
- [25] H. Haresamudram et al., "Masked reconstruction based self-supervision for human activity recognition," in *Proc. Int. Symp. Wearable Comput.*, Sep. 2020, pp. 45–49.
- [26] H. Xu, P. Zhou, R. Tan, M. Li, and G. Shen, "LIMU-BERT: Unleashing the potential of unlabeled data for IMU sensing applications," *GetMobile, Mobile Comput. Commun.*, vol. 26, no. 3, pp. 39–42, 2022.
- [27] E. Kim, S. Helal, and D. Cook, "Human activity recognition and pattern discovery," *IEEE Pervasive Comput.*, vol. 9, no. 1, pp. 48–53, Jan. 2010.
- [28] L. Chen, J. Hoey, C. D. Nugent, D. J. Cook, and Z. Yu, "Sensor-based activity recognition," *IEEE Trans. Syst., Man, Cybern. C, Appl. Rev.*, vol. 42, no. 6, pp. 790–808, Nov. 2012.
- [29] F. J. Ordóñez and D. Roggen, "Deep convolutional and LSTM recurrent neural networks for multimodal wearable activity recognition," *Sensors*, vol. 16, no. 1, p. 115, 2016.
- [30] J. Li, L. Yao, B. Li, X. Wang, and C. Sammut, "Multi-agent transformer networks for multimodal human activity recognition," in *Proc. 31st ACM Int. Conf. Inf. Knowl. Manage.*, 2022, pp. 1135–1145.
- [31] M. A. A. Al-Qaness, A. Dahou, M. A. Elaziz, and A. M. Helmi, "Multi-ResAtt: Multilevel residual network with attention for human activity recognition using wearable sensors," *IEEE Trans. Ind. Informat.*, vol. 19, no. 1, pp. 144–152, Jan. 2022.
- [32] H. Haresamudram, I. Essa, and T. Plötz, "Contrastive predictive coding for human activity recognition," *Proc. ACM Interact. Mobile Wearable Ubiquitous Technol.*, vol. 5, no. 2, pp. 1–26, 2021.
- [33] A. van den Oord, Y. Li, and O. Vinyals, "Representation learning with contrastive predictive coding," 2018, *arXiv:1807.03748*.
- [34] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
- [35] T. B. Brown et al., "Language models are few-shot learners," in *Proc. NIPS*, 2020, pp. 1877–1901.
- [36] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 16000–16009.
- [37] M. Joshi, D. Chen, Y. Liu, D. S. Weld, L. Zettlemoyer, and O. Levy, "SpanBERT: Improving pre-training by representing and predicting spans," *Trans. Assoc. Comput. Linguistics*, vol. 8, pp. 64–77, Dec. 2020.
- [38] M. Zhang and A. A. Sawchuk, "USC-HAD: A daily activity dataset for ubiquitous activity recognition using wearable sensors," in *Proc. ACM Conf. Ubiquitous Comput.*, Sep. 2012, pp. 1036–1043.
- [39] M. Malekzadeh, R. G. Clegg, A. Cavallaro, and H. Haddadi, "Protecting sensory data against sensitive inferences," in *Proc. 1st Workshop Privacy Design Distrib. Syst.*, 2018, pp. 1–6.
- [40] D. Anguita et al., "A public domain dataset for human activity recognition using smartphones," in *Proc. ESANN*, vol. 3, 2013, p. 3.
- [41] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
- [42] D. M. W. Powers, "Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation," 2020, *arXiv:2010.16061*.



Jinqiang Wang received the B.E. degree from Henan Normal University, Xinxiang, China, in 2020, and the M.E. degree from the University of South China, Hengyang, China, in 2023. He is currently pursuing the Ph.D. degree with the University of Science and Technology Beijing, Beijing, China.

His research interests include intelligent sensing and pattern recognition.



Wenxuan Cui received the B.E. degree from Changsha University of Science and Technology, Changsha, China, in 2022. He is currently pursuing the M.S. degree with the School of Computer Science, University of South China, Hengyang, China.

His research interests include intelligent perception and pattern recognition.



Huansheng Ning (Senior Member, IEEE) received the B.S. degree from Anhui University, Hefei, China, in 1996, and the Ph.D. degree from Beihang University, Beijing, China, in 2001.

He is currently a Professor with the School of Computer and Communication Engineering, University of Science and Technology Beijing, Beijing, and the Founder and the Principal with the Beijing Cyberspace International Science and Technology Cooperation Base, Beijing. His current research interests include the IoT, gen-

eral cyberspace and metaverse, smart education, cyber-syndrome, and cyber-health.



Tao Zhu (Member, IEEE) received the B.E. degree from Central South University, Changsha, China, in 2009, and the Ph.D. degree from the University of Science and Technology of China, Hefei, China, in 2015.

He is currently an Associate Professor with the University of South China, Hengyang, China. He is the principal investigator of several projects funded by the National Natural Science Foundation of China and the Science Foundation of Hunan Province. His research

interests include the IoT, pervasive computing, assisted living, and evolutionary computation.

Dr. Zhu is currently the Chair of IEEE CIS Smart World Technical Committee Task Force on "User-Centered Smart Systems."



Zhenyu Liu has published more than 20 academic papers in important domestic and foreign journals or international conferences, one of which was included in SEI. He has chaired or played a leading role in more than ten research projects funded by the National Natural Science Foundation of China, the Hunan Provincial Natural Science Foundation, the Hunan Provincial Science and Technology Plan, and the Hunan Provincial Department of Education Science.